

Total no. of pages:2

Roll No:

Second-SEMESTER B.Tech. (ECE/EIOT/EE/ICE)

MID-SEMESTER EXAMINATION, June 2022

Course Codes: ECECC02/EIECC02/EECC02/ICECC02 ✓

Course Title: Electronic Devices and Circuits

Time: 1:30 Hrs.

Max. Marks: 15

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed and mentioned in the answer.

Q. No.	Questions	Marks	CO
1. (a)	With an appropriate diagram of the (PNP or NPN) BJT, indicate and explain the various current components for the BJT operating in active mode.	1	CO1 & CO2
1. (b)	Explain the transfer characteristics and the output characteristics of the transistor in CE configuration.	2	
2. (a)	What are the various operational modes of a BJT with reference to the biasing of BE and CB junctions?	1	CO3
2. (b)	With a suitable example/diagram, explain the significance of the DC load line in deciding an appropriate Q-point of the transistor to be used as an amplifier.	2	
3. (a)	What are the various stability factors applicable to a DC bias circuit and what is meant by bias stabilisation?	1	CO3
3. (b)	Determine the operating point of the transistor in the circuit of Fig. 3, if $\beta=165$.	2	

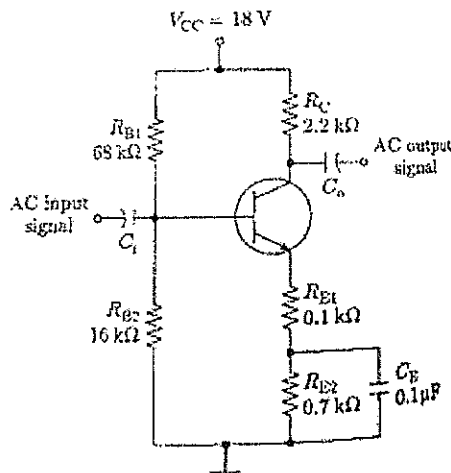
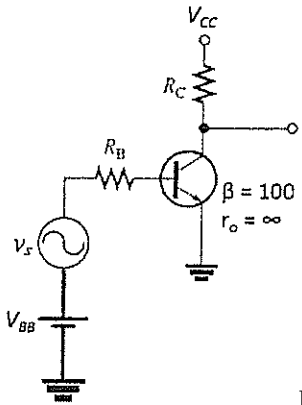
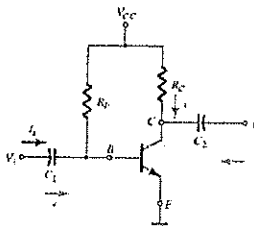


Fig.3

4. (a)	Based on their salient features, compare the CE and CC amplifier configurations of BJT,	1	CO3
4. (b)	Determine the voltage gain of the BJT amplifier circuit shown in Fig. 4, using small signal analysis based on hybrid- π model of the transistor. Given $V_{BB} = 3V$, $V_{CC} = 10V$, $R_B = 100\text{ k}\Omega$, $R_C = 3\text{ k}\Omega$.	2	
 Fig. 4			
5. (a)	Draw the small signal AC equivalent circuit of the common emitter fixed-bias circuit shown in Fig. 5, using r_e -model of the BJT.	1	CO4
 Fig. 5			
5. (b)	For the circuit shown in Fig. 5, drive the expressions for Z_i , Z_o , A_v , and A_i where symbols have their usual meaning.	2	